



Plan nacional de actuaciones preventivas de los efectos del exceso de temperatura sobre la salud

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Un verano de cuidado



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NATIONAL PLAN OF PREVENTIVE ACTIONS AGAINST THE EFFECTS OF EXCESSIVE TEMPERATURES ON HEALTH 2026 approved by the Interministerial Commission for the effective application of the National Plan of Preventive Actions on the Effects of Excessive Temperatures on Health at its meeting on April 28, 2026.

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Prologue

The association between high temperatures and increases in morbidity and mortality is very robust; numerous epidemiological studies show a significant increase in mortality above a certain thermal threshold, and it has been shown that epidemiological heat waves directly affect mortality from cardiovascular and respiratory diseases.

During the summer months of 2003, heat waves swept across Europe, causing a significant increase in morbidity and, as subsequently confirmed, mortality from causes where the heat waves were a contributing factor. This highlighted the significant impact of high temperatures on public health.

To prevent episodes like those that occurred that year, Spain launched the ***National Plan for Preventive Actions Against the Effects of Excessive Temperatures on Health in 2004***, with the aim of reducing the potential effects associated with epidemiological heat waves during the summer months. Since its implementation, the National Plan has been continuously improved over the years, generating a positive impact on the prevention of problems and illnesses related to exposure to epidemiological heat waves.

This is the ***twenty-second year*** that the Ministry of Health has planned and managed the National Plan, and its relevance as an instrument for preventing the effects of epidemiological heat waves on health is undeniable, since experts predict that heat waves will become more frequent and last longer due to climate change.

As every summer, the Ministry of Health makes available to health authorities and citizens information regarding the effects of heat waves
epidemiological data on health, as well as daily risk levels for heat waves
epidemiological monitoring during the summer period, with the aim of preventing its effects on health.

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Director General of Public Health and Health Equity

Background

During the summer months of 2003, a heat wave swept across Europe, causing a significant increase in morbidity and, as has since been confirmed, mortality from causes where the heat wave was a contributing factor. To prevent similar episodes, in 2004 the Ministry of Health launched the ***National Plan for Preventive Actions Against the Effects of Excessive Temperatures on Health***. Since then, the National Plan has been activated every summer from May to September.

The implementation of the National Plan is achieving its main objective: preventing health problems caused by heat waves. From a health perspective, exposure to high temperatures particularly affects children.

to older people and people with underlying chronic conditions. From a social point of view, marginalization, isolation, dependency, disability, and the living conditions of people with fewer resources add risk factors that make groups even more vulnerable, groups that, precisely because of their socioeconomic conditions, should be more supported.

It is important to mention that there is no universally accepted methodological definition of what constitutes a heat wave. From a climatological perspective, the Spanish State Meteorological Agency (AEMET) defines a heat wave as an episode of at least three consecutive days in which at least 10% of the weather stations considered record maximum temperatures above the 95th percentile of their series of daily maximum temperatures for the months of July and August during the period 1971-2000.

However, this definition does not take into account the health impacts of temperatures and, therefore, other approaches should be explored to ensure the effectiveness of preventive health plans for risk temperatures.

This is why an epidemiological definition of a heat wave or health risk is used. Epidemiological definitions of these events describe their onset threshold based on their health effects, generally using mortality as an indicator. Thus, this type of heat wave definition seeks out temperatures associated with mortality anomalies. Furthermore, given that

¹ Climatological definition of "heat wave" as an episode of at least three consecutive days, in which at least 10% of the stations considered record maximums above the 95th percentile of their series of daily maximum temperatures for the months of July and August of the period 1971-2000.

Specific episodes can have notable impacts on health; this definition of a heat wave is not conditioned on a specific duration of consecutive days.

The main advantage of defining heat waves in this way lies in its usefulness for public health alerts, as these are tailored to the actual vulnerability of the population to environmental exposure to extreme heat. Therefore, from now on, when this document refers to a "heat wave," it will be understood as an **epidemiological heat wave**.

The implementation of the National Plan is having a positive impact on the prevention of problems and diseases related to exposure to heat waves.

In general terms, the population has been informed about how to protect themselves and take care of those at greater risk, avoiding problems for the most vulnerable groups.

The design and development of meteorological and mortality information systems has been a key element in the Plan's success. One of the results worth highlighting is the implementation of a Mortality Information and Analysis System that allows for daily mortality monitoring, which is very useful for the rapid detection of other health problems.

The National Plan establishes measures to reduce the effects associated with excessive temperatures and to coordinate the relevant state institutions. It also proposes actions that can be implemented by the Autonomous Communities and Cities and by local authorities.

The National Plan establishes the actions planned for prevention and control, structured in several levels of action according to the level of risk reached as a consequence of the temperature increase. Furthermore, it proposes the collection of predictive information on ambient temperature and daily information on quantitative changes in mortality, and establishes the criteria for the information system that allows for the active monitoring of risks associated with exposure to

heat waves.

The mortality monitoring results obtained during the years the National Plan has been in effect also allow for the identification of excess mortality even before it exceeds the temperature-related risk thresholds established in successive summer campaigns. To this end, in 2026, as in previous summers, mechanisms will be implemented to establish early warning signals indicating a

excess mortality associated with heat waves.

An important aspect of this National Plan is its involvement with Social Services, since older people are the most vulnerable. The participation of administrations at the central, regional, and local levels, social organizations, and above all, awareness-raising and support for the most vulnerable groups, are crucial.

These measures are essential to minimize harm to the population. Another crucial element is providing information to citizens, high-risk groups, and healthcare and social services professionals.

The experience gained during the years of implementation of the National Plan since 2004 has allowed for the identification of changes necessary to improve its effectiveness. In 2015, an evaluation of the National Plan was carried out from 2004 to 2014. This made it possible to identify aspects that needed improvement to ensure the fulfillment of the Plan's health objectives. In this regard, information and communication to citizens was strengthened through the convergence of heat wave risk levels across the entire Spanish territory [1].

Along the same lines, and in order to optimize exposure to heat waves, as well as alert guidelines for the population, in 2017 it was agreed to launch a pilot plan whose geographical scope would be based on areas of meteosalud. The definition of the meteosalud zones was carried out through the zones of Weather forecasts with similar climatologies of Adverse Weather Phenomena (AWP) defined by AEMET [2]. This geographical disaggregation was established throughout the Spanish territory during the 2024 season, and remains in effect today.

Climate Change and Extreme Temperatures

During the last few decades, concern has increased about the effects of "Climate Change" which has materialized in the United Nations Framework Convention, the Kyoto Protocol and the agreement reached at the Paris summit (COP21, December 2015).

The latest findings from the working groups of the Intergovernmental Panel on Climate Change (IPCC) continue to demonstrate that the observed global warming due to climate change is unequivocal, that the impacts of climate change are already negatively affecting many physical and biological systems, and that these effects will increase. Furthermore, they state that it is indisputable that human activities contribute to climate change, making extreme events, such as heat waves, torrential rains, and droughts, more frequent and severe. Continued global warming caused by the increase in greenhouse gas concentrations in the atmosphere will have a broad and significant impact on the economy, the environment, and health. The projected effects of climate change are highly varied, affecting a wide range of ecological systems and socioeconomic sectors, and are unevenly distributed across different territories and regions. The Mediterranean region has

identified as one of the areas most vulnerable to climate change.

In order to improve and strengthen the capacity of the health sector to face the fight against climate change, the Ministry of Health and the Ministry for Ecological Transition and the Demographic Challenge have been developing instruments for analysis, diagnosis, evaluation, adaptation and monitoring of the impacts of climate change on health and the National Health System, evaluating scenarios and models, to help in decision-making, prioritize problems and propose adaptation and mitigation actions.

In light of current scientific knowledge, forecasts indicate an increase in the frequency and intensity of intense heat waves. Although there is currently no internationally agreed-upon definition of these events, it is generally accepted that this phenomenon is associated with abnormally high maximum and minimum temperatures for the time of year, as well as their persistence over time.

The Spanish territory presents significant geographical variability that must be taken into account when proposing and implementing health control and protection measures. The establishment of threshold temperatures and the assignment of levels

These are the basic elements for characterizing the heat wave phenomenon, and both elements have been established jointly and flexibly by the Ministry of Health and AEMET (Ministry for Ecological Transition and the Demographic Challenge), taking into account the contributions made by the Communities and

Autonomous Cities. The adaptation mechanisms of the population to high temperatures, as indicated by recent studies, make it necessary to periodically review these thresholds [3].

According to the projections of the sixth assessment report of the Intergovernmental Panel on Climate Change (IPCC), changes in the climate will occur throughout the 21st century, among which the following stand out at the European regional level:

- Temperatures will continue to rise at a faster rate than globally. • Extreme heat events will become more frequent, unlike those that will increase in frequency. extreme cold events.
- Sea levels will continue to rise at a similar rate to the global rate.

And specifically in the Mediterranean subregion (which includes all of Spain except

Canary Islands):

- An increase in aridity and forest fires is expected.
- An increase in extreme temperatures and a decrease in precipitation and decrease in snow cover.

The latest scientific evidence [3][5][6][7][8] reinforces the idea that prevention plans have a real effect on reducing mortality associated with heat waves, which together with the possibility of excessively hot summers recurring in our country justifies the continuation of the National Plan.

Regulatory framework and scope

Regulatory framework

[Order PRE/1518/2004](#), of May 28, which creates the Interministerial Commission for the effective application of the National Plan of Preventive Actions against the Effects of Excessive Temperatures on Health.

Plan activation period

In general, the National Plan is activated from **May 16th to September 30th** of each year.

Furthermore, a flexibility criterion is introduced that allows activation outside this period, through monitoring during the last fifteen days prior (May 1-15) and fifteen days following (October 1-15) the activation period of the National Plan mentioned above. Along with the established flexibility criteria, the National Plan will monitor maximum temperatures and health risk levels from heat waves in the event of unusual episodes of high temperatures occurring outside of the specified dates. previously mentioned.

Area of territorial disaggregation

The National Plan is broken down for **182 meteohealth zones**.

The definition of the meteohealth zones was carried out through the meteorological prediction zones with similar climatologies of Adverse Meteorological Phenomena (AMP) defined by AEMET [2].

On the other hand, in those Autonomous Communities and Cities that have their own Preventive Action Plans for excess temperature, other specific areas of disaggregation may be available for their geographical territories.

Health impact of heat waves

Health effects

Human exposure to high ambient temperatures can lead to an insufficient physiological response from the thermoregulatory system. Heat waves can disrupt our vital functions when the body is unable to compensate for variations in core temperature.

A very high temperature causes a loss of water and electrolytes that are necessary for the normal functioning of the different organs.

In some people with certain chronic diseases, undergoing certain medical treatments and with disabilities that limit their autonomy, thermoregulation mechanisms may become unbalanced.

Exposure to high temperatures can cause health problems such as cramps, dehydration, heatstroke, and heatstroke (with multi-organ failure that can include symptoms such as gait instability, seizures, and even coma). The heading identified as a direct cause of mortality due to excessive ambient temperature in the International Classification of Diseases and Causes of Death, 10th Revision, is **“X-30: Exposure to excessive natural heat.”**

The impact of exposure to high temperatures is influenced by physiological aging and underlying illnesses. A healthy individual typically tolerates a variation in their internal temperature of approximately 3°C without significant alterations to their physical and mental condition. Above 37°C, a physiological defense reaction occurs.

Older adults and children are more sensitive to these temperature changes. Older adults have a reduced sensation of heat and, therefore, a diminished ability to protect themselves. There is a correlation between a decreased perception of thirst and a reduced perception of heat, especially when they suffer from neurodegenerative diseases. This is compounded by the reduced thermolysis of the elderly (numerous sweat glands are fibrotic, and the capacity for capillary vasodilation is diminished). Reduced thermolysis capacity also occurs in diabetes and neurodegenerative diseases.

Children with chronic health conditions or those taking certain medications, like adults, may be more susceptible to heat-related illnesses. Furthermore, specific physiological characteristics in childhood, mostly related to body water composition, sweating patterns, and metabolic heat production, put infants at a thermoregulatory disadvantage compared to adults.

This is especially true when they exercise or are very physically active in hot or humid environments. Furthermore, children—and even adolescents—are unable or unwilling to take the necessary steps to prevent or replace fluid loss, and they expose themselves to direct sunlight to the point of exhaustion if not supervised.

adults. Those who are overweight or who wear too much clothing, especially if it doesn't breathe, are also more susceptible.

Impact on mortality

According to calculations for the 2000-2009 time series by the Carlos III Health Institute [10] The risk of mortality attributable to heat waves increases, with a 95% probability, between 9.1% and 10.7% for each degree that the ambient temperature rises above the health impact threshold for heat waves in the provinces of Spain. The mortality associated with heat waves in the referenced study was 13,119 deaths attributable to heat waves, that is, about 1,300 deaths per year. During the study period, 4,373 days of heat waves were recorded throughout Spain (bearing in mind that heat waves can occur simultaneously in several provinces on the same day and are counted differently); that is, each day with a heat wave episode, mortality increases, on average, by 3 deaths per day.

Excess mortality has been associated with periods of 3 or more consecutive days of unusually high temperatures, and its effects can be observed during such periods. or with a delay of up to three days.

According to estimates from MoMo (monitoring of daily mortality from all causes), deaths attributable to high temperatures between 2015 and 2025 totaled 27,564. In 2022, there were 4,789 deaths attributable to high temperatures.

Temperatures were the year with the most excesses, followed by 2025 with 3,832 deaths. These estimates vary between autonomous communities and provinces and are mainly concentrated among those over 75 years of age.

Risk factors

The main risk factors associated with exposure to heat waves are:

Personal factors

- People over 65 years of age.
- Infants and children under 4 years old.
- Pregnant women.
- People with cardiovascular, respiratory, and mental illnesses (dementia, Parkinson's, Alzheimer's...).
- People with chronic diseases (diabetes mellitus, morbid obesity...).
- People undergoing certain medical treatments (diuretics, neuroleptics, anticholinergics and tranquilizers).
- People with memory disorders, difficulties in understanding or orientation, or little autonomy in daily life.
- People with difficulties adapting to heat.
- People with acute illnesses during heat waves. • People who consume alcohol and other drugs.

Environmental, occupational, or social factors

- People who live alone. • Homeless people.
- People with unfavorable economic conditions.
- Living in homes that reach excessively high indoor temperatures due to a poor ability to regulate temperatures through active or passive means
- Exposure to heat waves for work-related reasons (manual work outdoors or requiring high contact with hot environments), sports-related reasons (sports of high physical intensity) or leisure activities.
- Exposure to heat waves in schools. • Environmental pollution.
- Highly urbanized environment.
- Prolonged exposure for several days to heat waves that last for the night.

Local factors

Although the above mechanisms act in a general way, local factors play a decisive role, since they condition the comfort temperature, the threshold temperatures to consider and the temperature-mortality association, that is, the magnitude of the impact.

The main local factors are:

- Demography, which determines the composition of the population pyramid and, therefore
Therefore, the importance of vulnerable groups.
- Climatology, insofar as individuals adapt to the local climate. This explains why the effect
of extreme temperatures does not depend on absolute values, but rather on whether
or not we are within the normal temperature range for a given location.

Plan Objectives

The **objective** of this National Plan is to reduce the impact on the health of the population as a result of heat waves.

The strategy is based on the following actions:

- Development of the Environmental Information System that includes the prediction of the temperatures.
- Providing information to the public about the health effects of heat waves and on protection and prevention measures.
- Development of the Mortality Information System.
- Information for healthcare and social services professionals.
- Coordination with social services for the identification of risk groups, especially in both children and very vulnerable people
older people.
- Coordination with the competent administrations and entities.

Coordination and management of the Plan

In order to guarantee the effective application of the National Plan, the Interministerial Commission will continue to perform the functions assigned in **Order PRE/1518/2004**, of May 28, which creates the Interministerial Commission for the effective application of the National Plan of Preventive Actions against the Effects of Excessive Temperatures on Health.

The Commission is chaired by the head of the Directorate General of Public Health and Health Equity of the Ministry of Health and is made up of one person

Representative of the following institutions with the rank of Deputy Director General:

- Ministry of the Interior. Directorate General of Civil Protection and Emergencies.
- Ministry for Ecological Transition and the Demographic Challenge. State Meteorological Agency (AEMET).
- Ministry of Social Rights, Consumer Affairs and Agenda 2030. Institute for the Elderly and Social Services (IMSERSO).
- Ministry of Territorial Policy and Democratic Memory. Directorate General of State Administration in the Territory.

The head of the General Sub-Directorate of Environmental Health will act as secretary and Occupational Health of the Ministry of Health.

The functions of this Commission are as follows:

1. Prepare the guidelines for compliance with the National Plan at the State level.
2. Establish the preventive and control strategies that are considered appropriate in light of the new evidence.
3. Activate the information levels in coordination with the Autonomous Communities and Cities and taking into account the frameworks of competences.
4. To propose the necessary organizational, structural and preventive measures to avoid or reduce the impact of thermal extremes on health.
5. Prepare risk assessment, management and communication plans.

The Interministerial Commission will request, when it deems necessary, the advice of Scientific Societies and public and private entities that it considers most appropriate for the performance of its functions.

The Interministerial Commission is the body responsible for coordination with public and private entities and has the necessary powers for the execution of This National Plan.

In this regard, coordination mechanisms will be established to guarantee the implementation of the National Plan with the following entities:

- Departments of Health and Social Services of the Autonomous Communities and Cities Autonomous.
- Ministries of the Interior (General Directorate of Civil Protection and Emergencies), for Ecological Transition and the Demographic Challenge (AEMET and General Directorate of Biodiversity, Forests and Desertification) and of Justice (Civil Registry).
- Public and private media.

Methodological criteria applied

For the proper implementation of the National Plan, AEMET provides daily maximum temperatures expected for that day and forecasts for the following two days. These temperatures are provided broken down by two population-based reference weather stations for each weather and health zone.

Risk levels will be assessed to determine when to notify the affected Autonomous Community or Autonomous City, in order to implement the measures provided for at the corresponding level together with them.

Reference threshold temperatures for health impact from heat waves by meteohealth zones

The Strategic Plan for Health and Environment 2022-2026 (PESMA) of the Ministry of Health establishes among its actions to be developed in the framework of heat waves under The research intervention line, the characterization and evaluation of weather health zones, as well as the unification of alerts, were carried out during the 2024 season. Work was done to establish the weather health zones, with the aim of making the information available to the stakeholders involved in the National Plan and the general public.

Threshold temperatures for health impacts from epidemiological heat waves were calculated by analyzing the association between time series data on mortality and temperatures at the level of meteohealth zones. The meteohealth zones were defined using meteorological prediction zones with similar climatologies of Adverse Weather Phenomena (AWP) as defined by AEMET (the Spanish State Meteorological Agency). This procedure is based on the methodology developed and patented by the National School of Public Health of the Carlos III Institute of Health.

The variables used for the determination of threshold temperatures are They established in:

- The dependent variable used was the daily death count from natural causes (ICD-10: A00-R99) occurring in the meteohealth zones. This data was provided by the National Institute of Statistics (INE), at the municipal level, for all municipalities in the country.
- The independent variable was the maximum daily temperature in degrees Celsius.
This temperature was established as the average of the data recorded in two

² Determination of epidemiological temperature thresholds for defining heat waves and cold waves based on daily population mortality. Julio Díaz, José Antonio López-Bueno, Cristina Linares. General Registry of Intellectual Property. Registration Number 16/2023/2287. April 2023.

Automatic weather stations were selected based on the population they represent within the meteosalud area and agreed upon with the corresponding regional health authority. The data were provided by AEMET.

The time series used includes all daily cases between January 1, 2012 and December 31, 2023; with the exception of the years 2020 and 2021, since, after a descriptive exploration of the data, these were discarded due to significant mortality anomalies, probably related to the impact of the COVID-19 pandemic in Spain.

Some weather and health zones were excluded due to an excessive number of missing values in the temperature records (>10% missing values). When these zones had a lower proportion of missing values, the missing values were imputed by linear interpolation or by predicting a linear model controlled for seasonality, depending on whether they were isolated missing observations or longer gaps.

To determine the temperature threshold for an epidemiological heat wave or cold wave in each weather zone, a statistical analysis of the association between mortality and temperature was performed, as recommended by the World Health Organization (WHO). First, Box-Jenkins models for mortality in each weather zone were fitted, controlling for seasonality and trend variables (*Autoregressive Integrated Moving Average family models*). These models, therefore, represent the expected behavior of the series based solely on historical data. Anomalous mortality values due to external variables not considered, such as temperature, will cause a model mismatch, resulting in an increased residual (the difference between the expected and predicted values). It is these residuals from the Box-Jenkins models for daily mortality during the summer months (June-September) that will be represented and analyzed. Specifically, the temperature points statistically associated with abnormally high residuals will be identified.

This indicates the association between temperature and abnormally high manifestations of summer mortality that are not predictable based on the usual time series behavior of mortality.

These temperature points associated with mortality anomalies are established using Cartesian diagrams. In these diagrams, the model residuals are plotted on the y-axis, and the maximum daily temperature on the x-axis.

A horizontal line marks the upper confidence limit at the 95% probability of the mean of the model errors. It is taken into account that the effects of heat waves on mortality occur over a prolonged period, with an initial peak in mortality during the first three days of the event; therefore, a moving average was calculated to determine the temperature thresholds for epidemiological heat waves.

of the residual errors in 3 days.

Next, the average model error was calculated for temperature ranges of $\pm 1^{\circ}\text{C}$, where i represents each degree Celsius within the maximum temperature range. In other words, the average model error is calculated for temperature intervals of two degrees each, covering the entire temperature range and overlapping with each other.

These average residual values for temperature intervals are represented on Cartesian diagrams along with their two-sided 95% confidence intervals. Thus, the threshold temperature (Threshold) for defining a heat wave is established as that which is associated with errors statistically greater than the mean error of the model. Once the threshold temperature is determined in degrees Celsius, these cannot be directly compared to each other since they depend on local climatology. Therefore, they were translated into threshold temperature percentiles.

Since mortality anomalies are not always observed in a clearly defined manner, a series of criteria have been established to determine the threshold temperature in such cases:

- As a general criterion, the lowest temperature at which the first anomaly in the analyzed mortality is detected will be considered.
- If the first anomaly in mortality is below the 85th percentile, the following circumstances will be taken into account:

The initially detected threshold temperature is correct, although it may shift slightly because there are several "spikes" of anomalies at higher temperatures.

The detected threshold temperature is correct, but the sustained anomaly over time is not clearly visible, which may be due to the low mortality rate observed. In this case, the provincial percentile for that area and the temperature corresponding to that percentile are assigned.

Or if the temperature anomaly is clearly defined, the temperature obtained.

- If the first mortality anomaly is above the 99th percentile, the provincial percentile for that area and the temperature corresponding to that percentile will be assigned.
- In those cases where the threshold temperature is not conclusive, because several temperatures with anomalies are detected in the analyzed mortality, the threshold temperature will be established by consensus.

For areas that showed an excess of missing values in the temperature records ($>10\%$ missing values), it was agreed to take as the threshold temperature of that meteohealth area, the percentile of the province obtained from the time series of its meteorological mirror area defined by AEMET (Annex IV).

In the methodology described, two tools were used: the free software R.4.4.2. and the IBM SPSS Statistics 28.0.1.1 software (15).

Risk levels derived from heat waves

Selection of limits for the different levels of health risk from heat waves

Once the threshold temperatures for health impact from heat waves have been calculated In each province, different alert scenarios were simulated based on the operation of the decision algorithm. This algorithm quantifies the heat wave index as the number of degrees by which the temperature threshold is exceeded, according to AEMET's forecasts at the reference observatories established for each day and the two following consecutive days.

Thus, heat wave indices were calculated for a random sample of provinces during the period 2009-2019. The provinces were recruited for the sample by generating an equiprobable random sequence of ones and zeros. As a result, the following 24 provinces were included: Alicante, Asturias, Ávila, Balearic Islands, Barcelona, Cádiz, Castellón, Córdoba, Girona, Granada, Huelva, Huesca, Madrid, Murcia, Ourense, Pontevedra, Salamanca, Santa Cruz de Tenerife, Segovia, Teruel, Toledo, Valencia, Valladolid, and Zaragoza.

For each of them, the heat wave indices, their 90th percentiles, and the median of the 90th percentiles in the 24 provinces were calculated – 90th percentile of the heat wave index = 3. From there, different alert scenarios were explored with the following transitions from yellow to orange (first limit) and orange to red (second limit):

Table 1. Scenarios explored for establishing health risk levels for heat waves

Scenarios	First limit	Second limit
E1	2	4
E2	2,5	4
E3	3,5	5
E4	3,5	7
E5	4	7

Finally, the values from the fourth transit scenario were taken, as they establish a more proportionate escalation in the progression of the alert level.

This represents a fixed transition in all provinces, with the inherent limitations of applying universal thresholds in provinces and areas that are essentially different in terms of their climate behavior and vulnerability status to heat waves.

However, this transition between risk levels will be applied provisionally until the specific risk factors for each province, as well as the forecast date, are established. At that time, a system based on the direct quantification of estimated risks will be implemented; these risks are currently considered indirectly, as higher-index episodes are associated with greater health impacts.

intensity.

Health risk levels from heat waves

The criteria for assigning health risk levels for heat wave situations are

It is based on a decision algorithm that relies on:

- The difference between the maximum expected temperature and the threshold temperature (only when the maximum expected temperature is higher than the established threshold temperature), with a persistence over time of 3 days.
- The resulting value will be multiplied by a “risk factor”³ depending on the weather health zone.
- Finally, the resulting value from the three days is added to the result obtained will decide the level of risk.

Equation 1. Alert level decision algorithm

$$\ddot{y}(\text{on in } 1 \ddot{y}) \ddot{y} \text{ in } 1 \ddot{y} + \ddot{y}(\text{on in } 2 \ddot{y}) \ddot{y} \text{ in } 2 \ddot{y} + \ddot{y}(\text{on in } 3 \ddot{y}) \ddot{y} \text{ in } 3 \ddot{y}$$

The assignment of health risk levels (*Table 2*) is carried out using the following criteria based on the value obtained in the decision algorithm:

- If the result obtained in the decision algorithm is 0, the index is “0”, the assigned level is called “**Level 0**” or absence of risk, and is represented by the **color green**.
- If the result obtained in the decision algorithm is greater than 0 and less than or equal to 3.5, the index is “1”, the assigned level is called “**Level 1**” or low risk, and is represented by the **color yellow**.
- If the result obtained in the decision algorithm is greater than 3.5 and less than or equal to 7, the index is “2”, the assigned level is called “**Level 2**” or medium risk, and is represented by the **color orange**.
- If the result obtained in the decision algorithm is greater than 7, the index is “3”, the assigned level is called “**Level 3**” or high risk, and is represented by the **color red**.

3. For this year, the risk factor will be 1 for all days and all weather zones. This factor will be modified in subsequent years, futures adapting it to the circumstances of each territory.

Table 2. Definition of health risk levels for heat wave

Risk Level	Denomination	Index
0	Absence of risk	0
1	Low risk	1
2	Medium risk	2
3	High risk	3

Information and surveillance systems

Current health surveillance systems have adequate information systems to know in advance the risk of temperature increases with acceptable reliability, as well as the impact that these temperatures have on the health of the population measured in terms of mortality.

System objectives

- To know in advance the health risk of heat waves that may affect a specific population residing in a particular geographical area.
- To understand the real impact of heat waves on public health.
- To identify the need to strengthen available healthcare resources.

Features

Within the institutional framework, it is a cooperative system involving different areas of health and non-health administration (AEMET, Ministry of Health, IMSERSO, Ministry of Justice, Autonomous Communities, ISCIII, etc.)

Different levels of information are established, according to the risk, based on the background and characteristics of each territory, the available time series and the available temperature predictions.

Environmental information system

Preventing the health effects of heat waves is largely possible. AEMET (the Spanish State Meteorological Agency) is currently able to predict maximum temperatures with high reliability and several days in advance.

Meteorological variables

The meteorological variables taken into account are: the maximum temperatures forecast for 3 days and the reference thresholds for health impact due to heat waves.

This information is provided daily by AEMET, broken down by weather and health zones, and sent electronically. This information, also sent daily and electronically, along with the heat wave health risk map, is distributed to the health authorities of the Autonomous Communities and Cities.

as well as the rest of the institutions involved in the National Plan.

Daily Mortality Monitoring (MoMo)

Within the framework of this National Plan, the monitoring of daily mortality is complementary to meteorological information. On the one hand, it allows for evaluation

situations of risk to health, assess the impact of heat waves on mortality and identify excess mortality from all causes.

The objective of mortality monitoring is to improve prevention and response capacity. MoMo and the Kairós Index are daily mortality surveillance systems associated with heat waves, developed by the Daily Mortality Surveillance Unit (MoMo) of the National Center for Epidemiology (CNE) at the Carlos III Health Institute, and contribute to the Plan. The Kairós Index, implemented in the summer of 2021, provides mortality alerts associated with heat waves, and since April 2022, MoMo has used a new model that, in addition to excess mortality from all causes, estimates the impact of the heat wave on population mortality, providing estimates of excess mortality attributable to the heat wave. The CNE of the Carlos III Health Institute

Carlos III Health will inform the Ministry of Health daily of any warning signs detected, according to the criteria defined in models^{4,5}.

Daily information and reports generated by this model during the activation period of the National Plan are available to the members of the Interministerial Commission.

Methodology

MoMo and the Kairos Index use three data sources:

- Daily deaths for the last ten years, excluding those from 2020.
This series is obtained from consolidated data from the National Institute of Statistics (INE), as well as the most recent deaths provided by the Ministry of Justice on deaths from all causes reported in civil registries of computerized municipalities (which corresponds to approximately 93% of the population).
- Temperatures, according to AEMET at the provincial level, in the same period of time, and including the current year.
- Population by age group and province, extracted from the INE.

The analysis is performed both for the overall population and by age groups. The age groups used are: 0-14, 15-44, 45-65, 65-74, 75-84, and over 85 years.

As an alert system, the Kairos Index establishes daily warnings that define different levels of mortality risk. There are three levels of the Kairos Index: Level 1, 2, and 3, which define low or no, moderate, and high mortality risks, respectively. These warnings apply to the current day and the following four days, as well as for...

⁴ https://momo.isciii.es/panel_momo/

⁵ <http://momo.isciii.es/kairos/>

population scope (national, of Autonomous Communities and Cities (CCAA) and provincial) and by age group (all ages, 0-14, 15-44, 45-65, 65-74, 75-84, plus 85 years).

As a system for estimating the impact of heat waves on population mortality, MoMo provides daily estimates of the number of deaths from all causes associated with heat waves for the current day for each population area and age group mentioned. The Kairós Index, which indicates the probability of excess mortality attributable to heat waves, will be published daily at the national, regional, and provincial levels on the website of the Carlos III Health Institute.

The Kairós and MoMo Index uses a GAM (*generalized additive model*) Poisson model, which is adjusted by province, level for which daily temperature information is available, and by age groups: 0-14, 15-44, 45-65, 65-74, 75-84, plus 85. The models include a term for trend, seasonality and for the variables ATO (*accumulated thermal overcharge*), defined as the number of degrees that the maximum temperature is above the mortality trigger threshold by province, multiplied by the number of days and FATO defined as the number of degrees that the minimum temperature is below the mortality trigger threshold, multiplied by the number of days⁷.

Mortality risk alerts

The model detects situations of excess mortality (alerts) due to heat waves. To do this, it measures the probability of a 10% or greater increase in the mortality rate attributable to heat waves.

The Kairos Index is constructed according to the probability of excess obtained:

- **Kairos 1:** If the probability of excess is less than 40%. Indicates the absence of mortality warnings attributable to the heat wave.
- **Kairos 2:** If the probability of excess is between 40% and 60%, it indicates a warning of mortality attributable to the medium-level heat wave.
- **Kairos 3:** If the probability of excess is above 60%, it indicates a warning of mortality attributable to the high-level heat wave.

⁶ <http://momo.isciii.es/kairos/>

⁷ https://momo.isciii.es/panel_momo/#section-documentación

Performances

All daily reports from MoMo and the Kairós Index, which include alerts of excess mortality at the national level, will be published daily on the website of the Carlos III Health Institute.

Information about MoMo and the Kairós Index will be published weekly on the social media channels of the Carlos III Health Institute.

Deaths from heat stroke

Additionally, information is monitored on

Mortality due to heat stroke, in this way when a death occurs associated with exposure to the heat wave, the health authorities of the Autonomous Communities must complete the information collected in Annex II and send it to the General Subdirectorate of Environmental Health and Occupational Health of the Ministry of Health.

The aim is to improve the monitoring of heatstroke mortality.
during the activation period of the National Plan.

Program for the care of the most vulnerable people

Through the municipal social services network, responsible for managing services such as home care, telecare and social centers, work will be done to identify the target population, since the first two services are aimed at vulnerable groups.

The distribution of information for protection and prevention through this network will aim to reach the most vulnerable population. To this end, collaboration agreements may be established with the Spanish Federation of Municipalities and Provinces to reach local councils.

The activation of the remaining specialized resources (day centers, residences, housing, occupational centers, etc.) will be coordinated through the Autonomous Communities and Cities.

The identification and care of the most vulnerable people will be carried out by the services available in the health and social care networks.

Preventive actions associated with each level of risk

The following describes the preventive actions associated with each level of health risk due to a heat wave.

Preventive actions associated with Levels 0 and 1 of health risk due to heat wave

The actions described below refer to those that are carried out systematically for the activation and implementation of the National Plan and that are maintained during periods of time when risk levels are 0 and 1.

- Inform the health authorities of the Autonomous Communities and Cities about the launch of the National Plan in its annual campaign on May 16, as well as the request for information regarding emergencies, admissions and deaths attributed to heat stroke or the effects of the heat wave.
- Implementation of the action programs by the agreements of social services.
- Informing the population about the existence of the National Plan and its compatibility with those of the Autonomous Communities and Cities that already have one.
- Information for the public on general protection and prevention measures individuals and in the immediate environment.
- Information for the population about the most vulnerable people and groups • Information for the population about the meaning of risk levels.
- Daily distribution via electronic means by the Department, to the members of the Interministerial Commission, to the ministerial departments integrated into it, to each of the representatives communicated by the Autonomous Communities and Cities and to the authorities of the Ministry of Health of the maps by zones of meteohealth of risk levels for heat wave and of the information on which they are based.
- Daily incorporation into the Department's website of information on heat wave risk levels, broken down by Autonomous Communities and Cities, activating hyperlinks with their own information systems, for those Autonomous Communities that have them.
- Updating and maintaining authority directories
Public Administrations involved in the Plan.
- Coordination with the health authorities of the Communities and Cities
Self-employed women who develop and implement their own Plan.
- Coordination with the Autonomous Communities and Cities for the implementation of the information campaign for professionals and public awareness campaign.

- Monitoring of information related to activities related to social policy.
- Implementation of the Mortality Health Information and Surveillance System attributable to the heat waves.
- Preparation of periodic statistical summaries of meteorological and socio-health information generated by the various Systems Information.

Preventive actions associated with Level 2 health risk due to heat wave

In addition to maintaining the measures planned for the two previous levels, the following are also contemplated:

- Immediate communication to the members of the Interministerial Commission of the presentation of the level through electronic means (SMS).
- Immediate communication to those responsible in the Communities and Cities Self-employed women affected by electronic means (SMS).
- Intensification of coordination with the Autonomous Communities and Cities to information for healthcare and social services professionals.
- Increased information on the meaning of the level, and issuance of advice and health recommendations appropriate to the level.
- Increased information dissemination to the public about the most vulnerable individuals and groups vulnerable to exposure to heat waves.
- Assessment of the adoption of additional measures of a general nature or aimed at specific groups.

Preventive actions associated with Level 3 health risk due to heat wave

- Reinforcement of the measures applied at Level 2.
- Intensification of information on the meaning of the level, and issuance of health advice and recommendations to the at-risk population.
- Immediate communication to the members of the Interministerial Commission and to those responsible for the Autonomous Communities and Cities of the presentation of the level, for the assessment of the actions that may be necessary in each case to manage the risk.
- Assessment by the Center for Coordination of Health Alerts and Emergencies of state of the situation.

Risk communication

Communication and information to the population, high-risk groups, health professionals and social services

During the activation period of the National Plan, all relevant information (health risk level, general recommendations, preventive actions, etc.) will be available on the Ministry of Health's website. Direct access will also be provided on the AEMET website⁸, and citizens will be informed through social media such as Facebook and X.

A "Temperature and Risk Level Information Service" will be made available to those citizens who wish to use it, through which they can receive an email and/or SMS with daily information on temperatures and the health risk level of those meteohealth zones that are of interest to them.

The estimate of excess deaths attributable to the heat wave in the MoMo panel, as well as the Kairos levels of mortality risk in the Kairos panel, will be made available to those citizens who wish to do so; they will be able to select territorial scope, sex and age group in each observation.

Information will be provided to the media to offer helpful advice and practical measures for preventing the health effects of heat waves. These awareness-raising activities will aim to increase individual prevention capacity and enable people to cope with heat waves by applying simple and accessible measures. A key objective is risk prevention and anticipation. To this end, efforts will be made to foster solidarity and prevention capacity within families, neighborhoods, and communities, especially to support those who are ill and most socioeconomically vulnerable.

Measures already established at the national level in the area of social services will be strengthened so that people at risk, their families, neighbors, etc. can report emergency situations or receive information and mobilization of help if necessary.

⁸ <https://www.aemet.es/es/portada>

Plan Monitoring

Weekly summaries

During the activation period of the National Plan, daily and weekly monitoring is carried out, detailing the health risk levels issued up to the time of the end-of-season report.

Plan Evaluation

Once the summer season is over, a final report will be prepared summarizing the current season.

This report will include information on:

- Weather conditions in Spain during the summer months • Activation of risk levels
- Daily Mortality Monitoring (MoMo)
- Specific mortality
 - or ICD10 – X30 (Exposure to excessive natural heat) or Number of deaths reported to the Ministry of Health during the activation period of the National Plan by the health authorities of the Autonomous Communities and Cities
- Information for the public

This report is published on the website of the Ministry of Health, where reports from 2004 onwards can be consulted.

Indicators

The following are established as indicators for monitoring the National Plan:

- Number of high temperature alerts issued during the period of activation of the Plan
- Percentage of days on alert for high temperatures throughout the campaign
- Total number of days on alert for high temperatures/Total number of campaign days
- Number of emails sent to citizens through the information service
- Number of SMS messages sent to citizens through the information service
- Mortality from selected causes during the activation period of the Plan
 - National
 - or ICD10-X30 (Exposure to excessive natural heat)
 - or Number of deaths reported to the Ministry of Health during the activation period of the National Plan by the health authorities of the Autonomous Communities and Cities

- Estimate of excess deaths attributable to excess temperatures (MoMo) by age, sex and territorial scope.
- Kairos levels of mortality risk by age, sex and territorial area.

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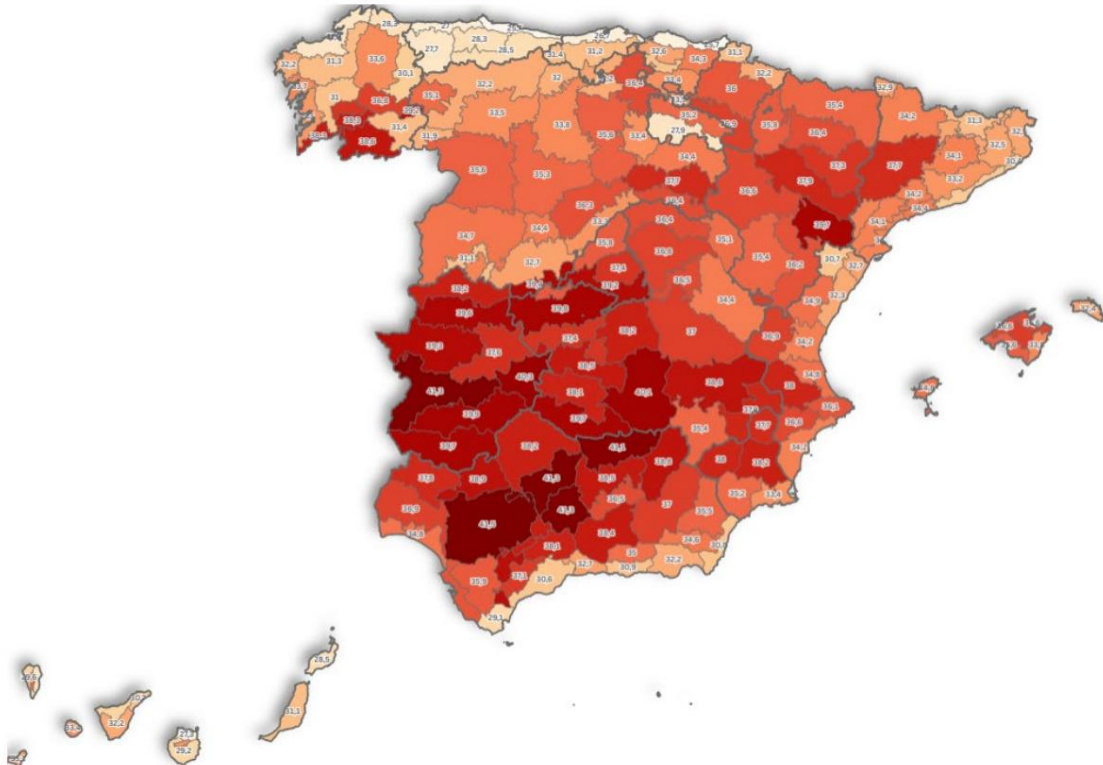
Attachments

Annex I. Reference thresholds by meteohealth zones of health impact from heat waves

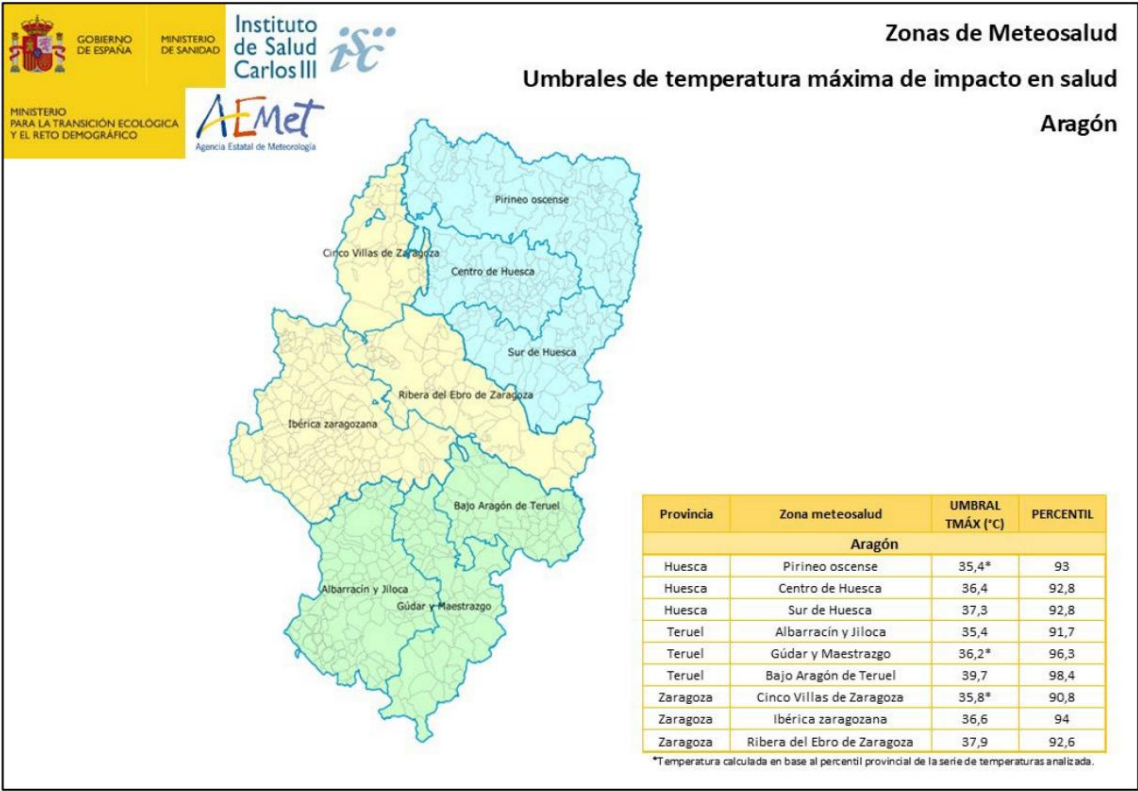
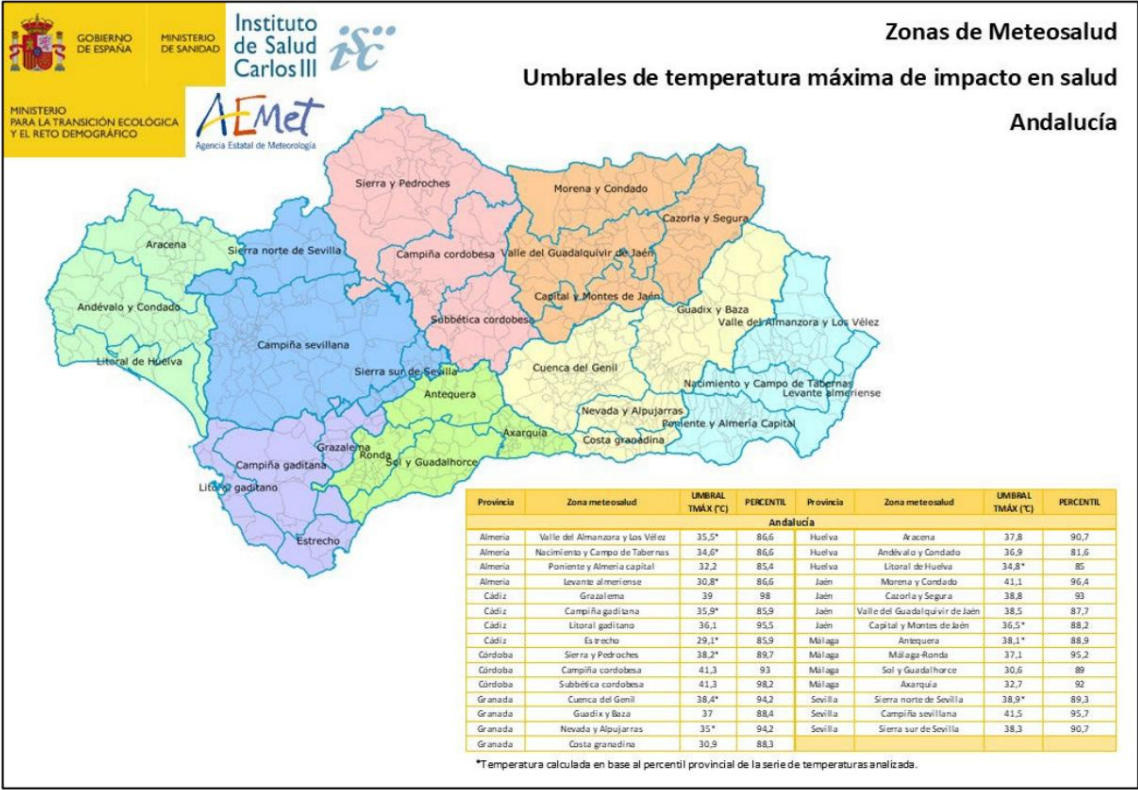


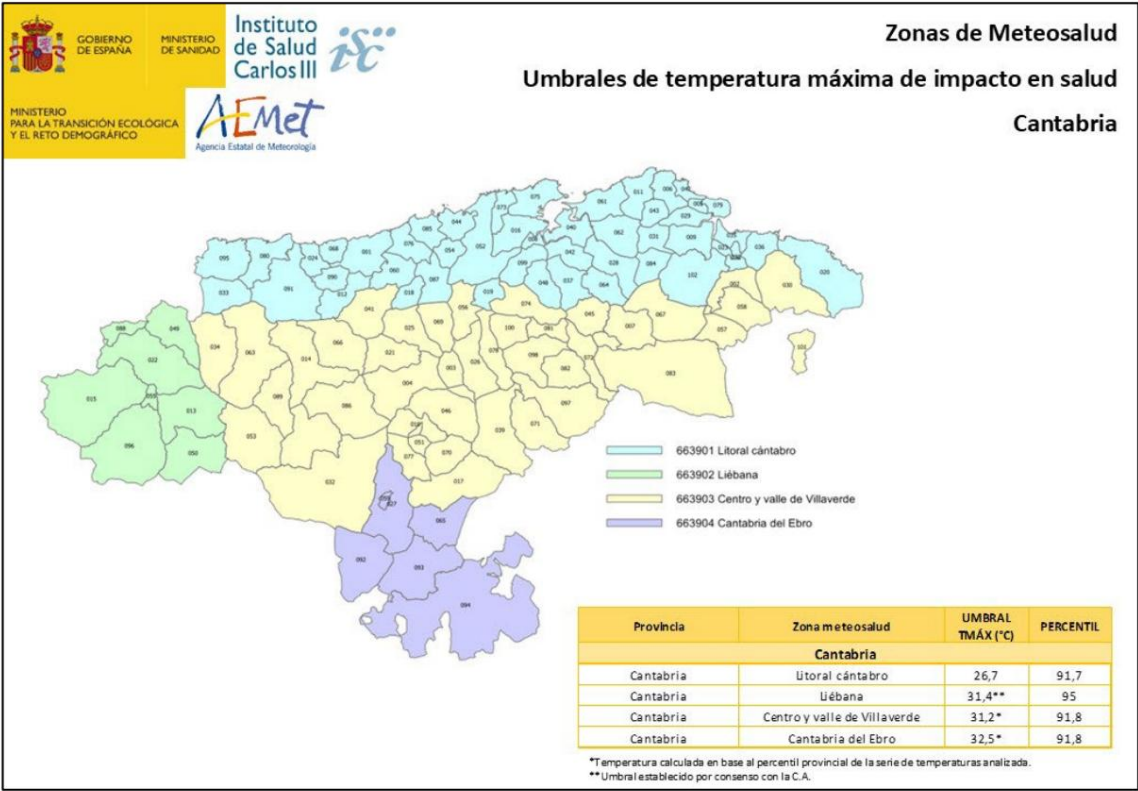
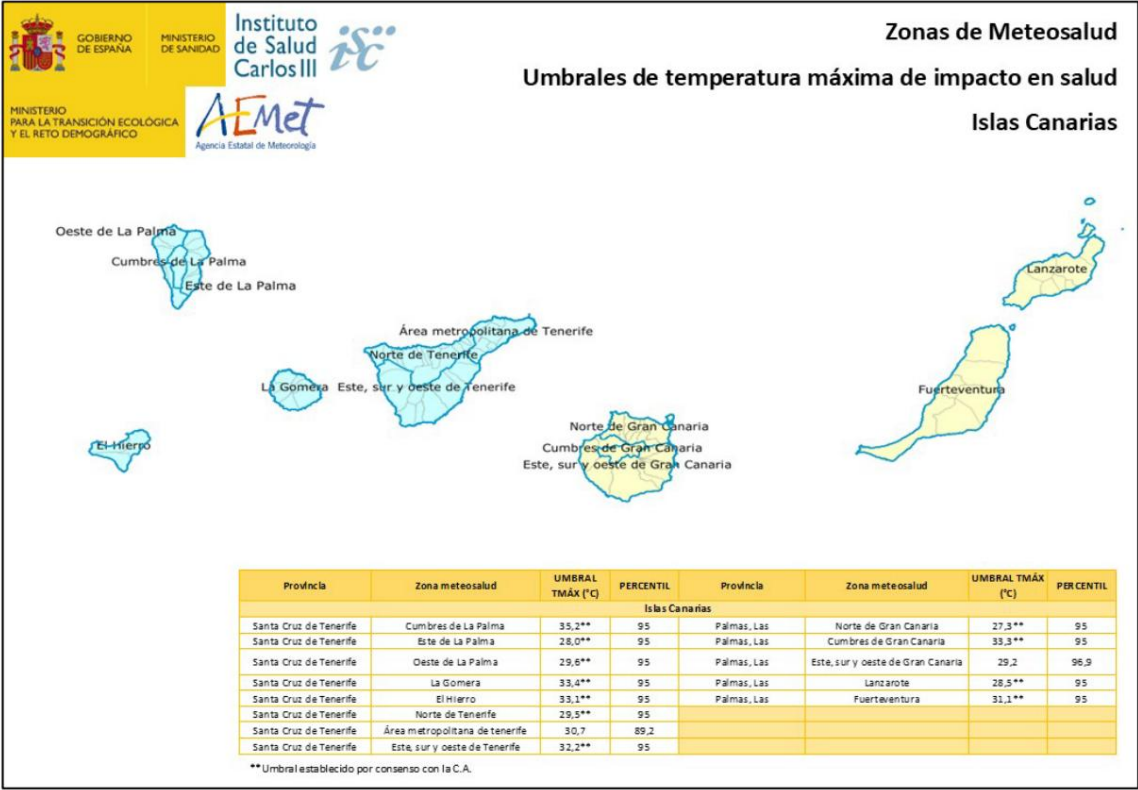
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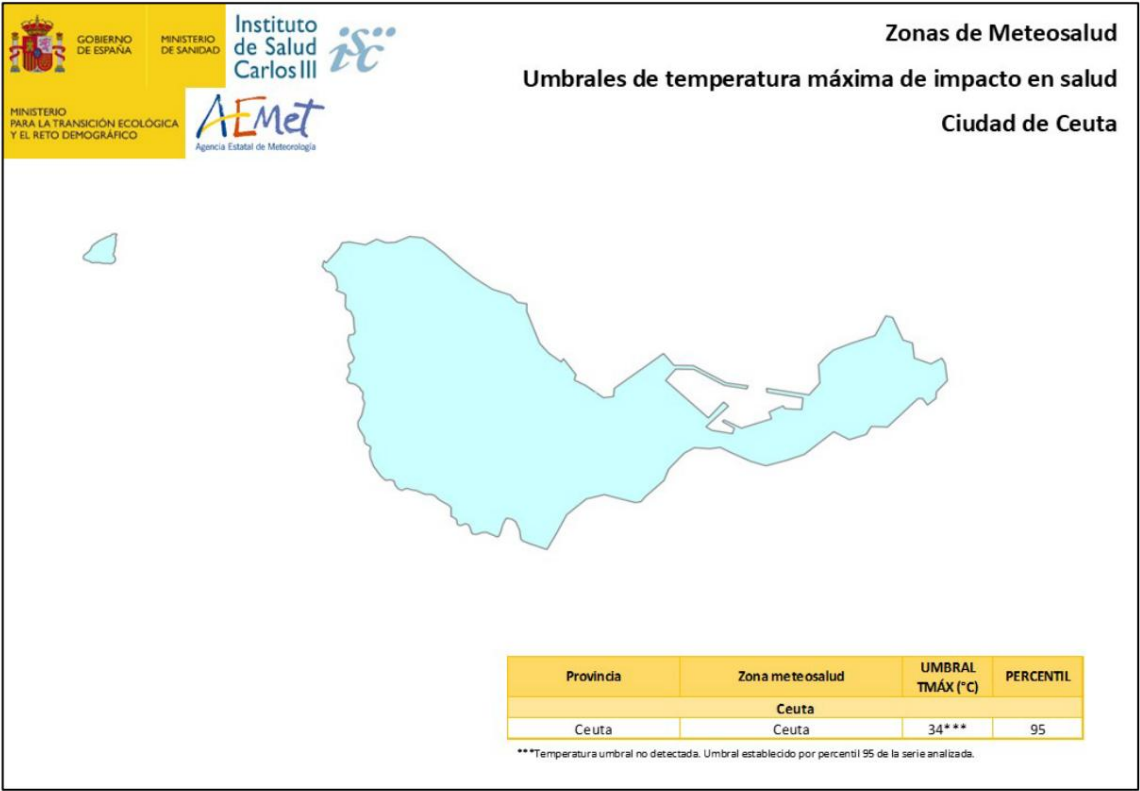
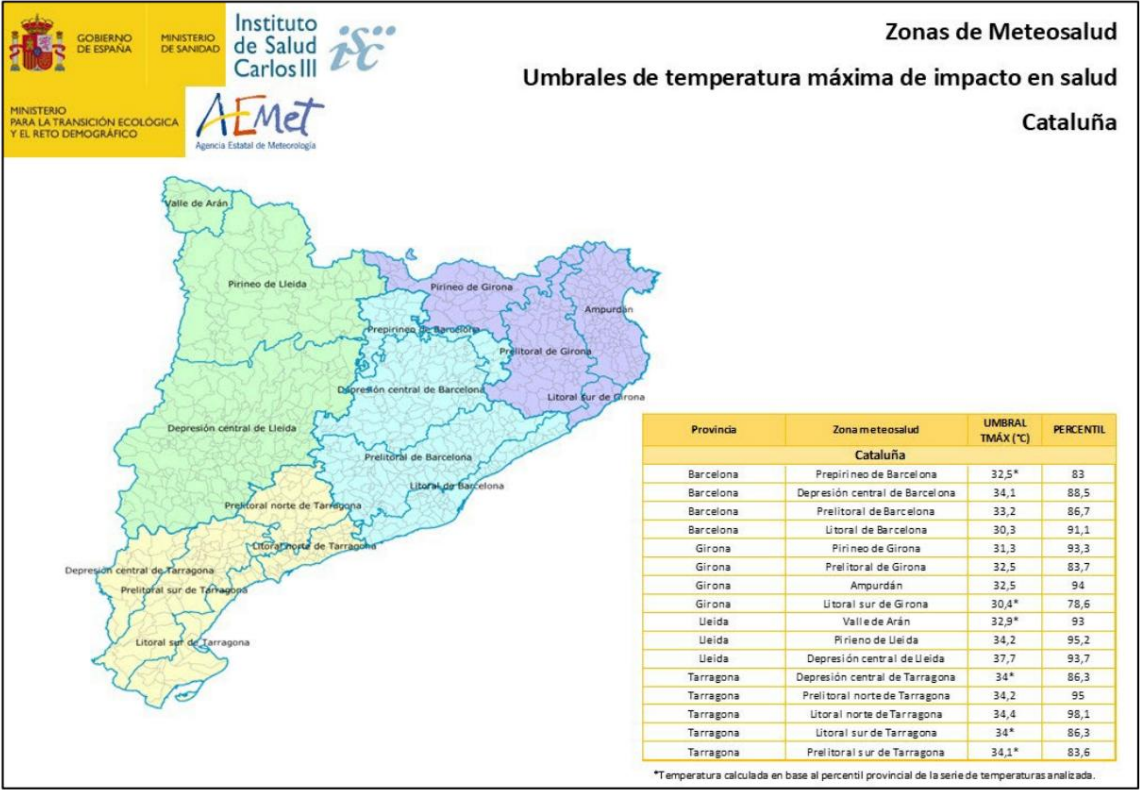
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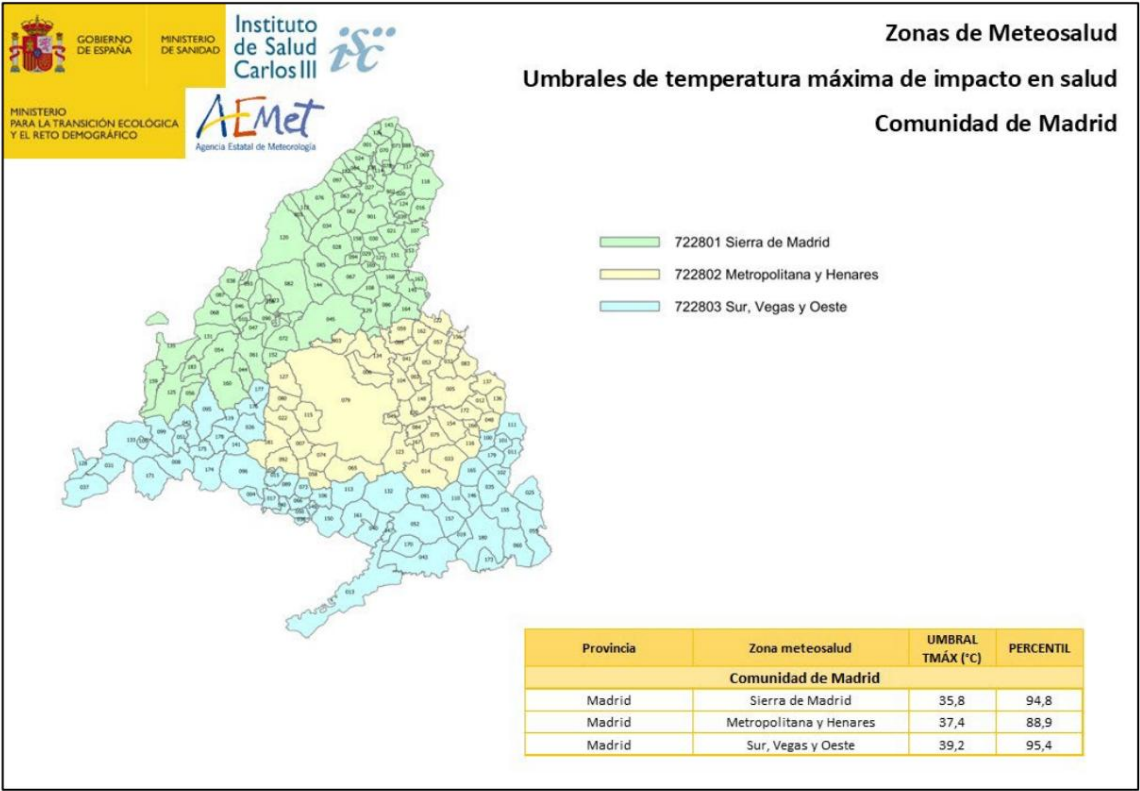
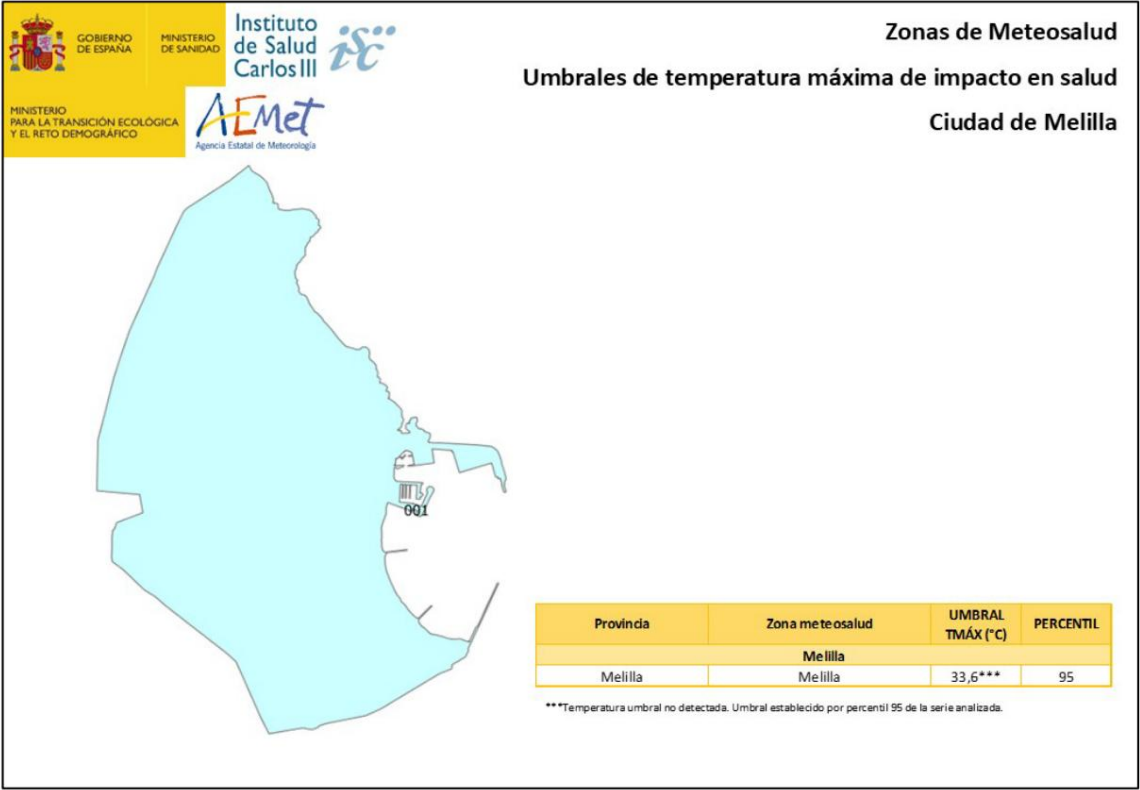


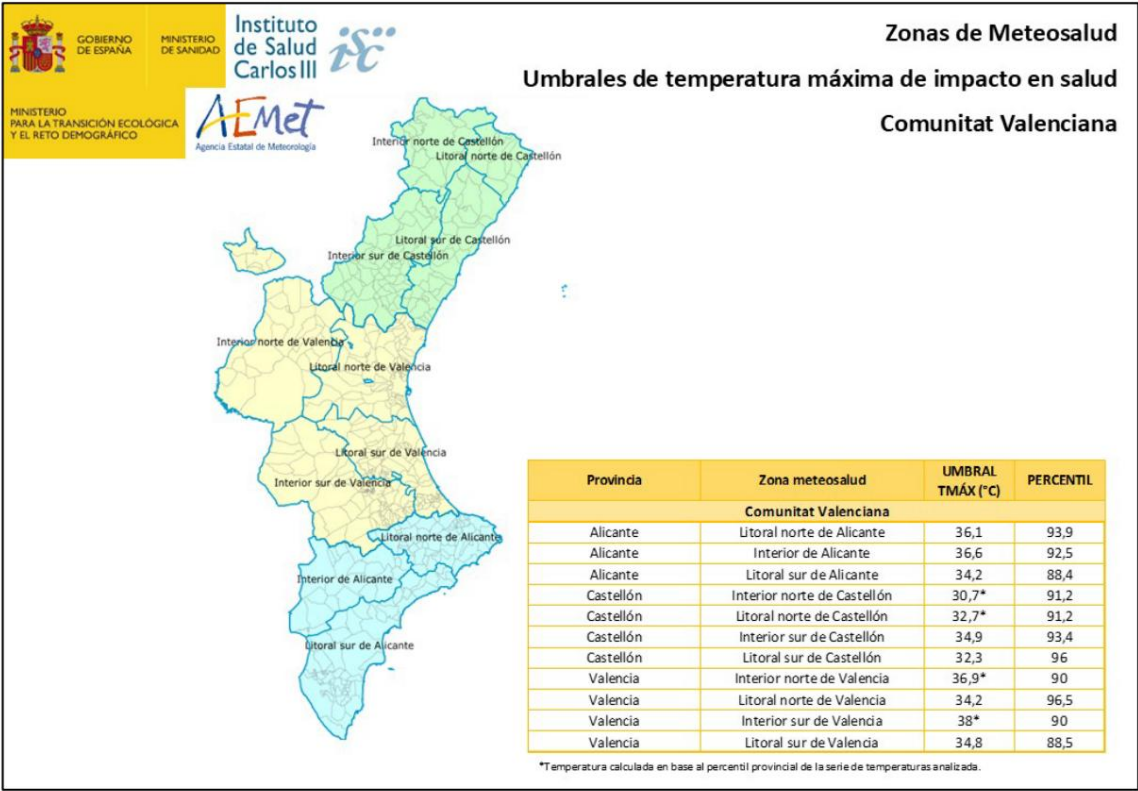
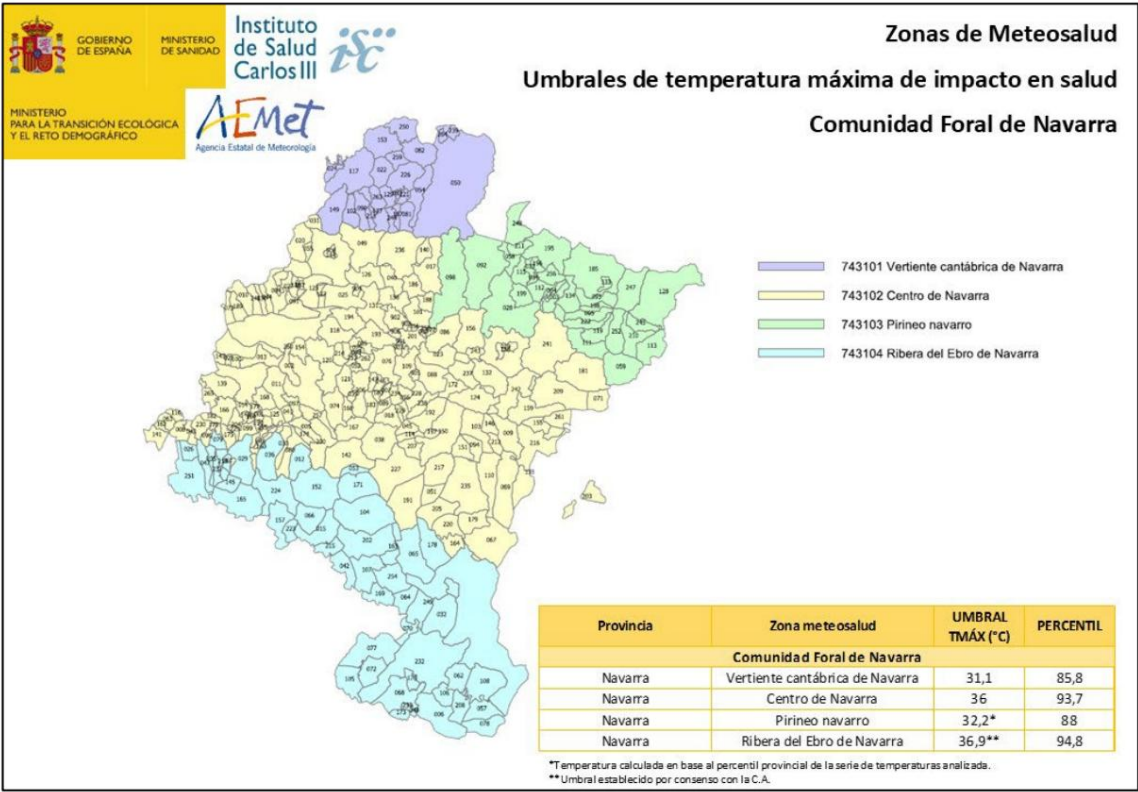
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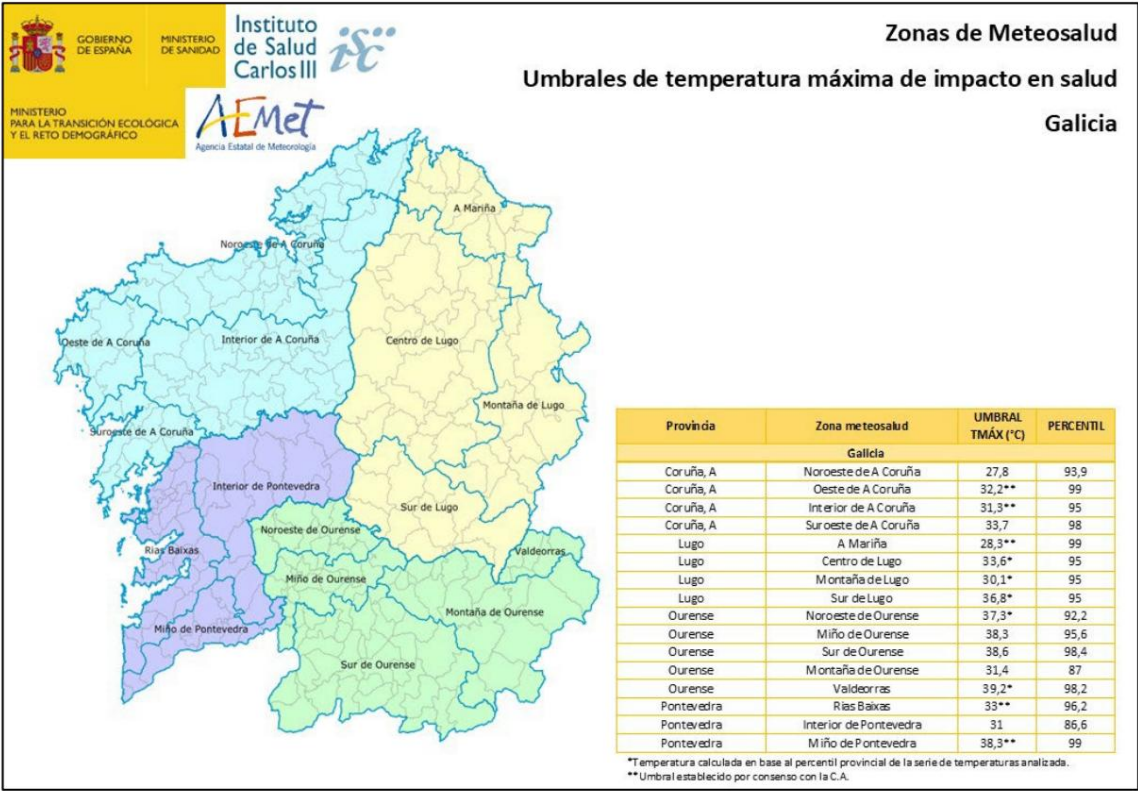
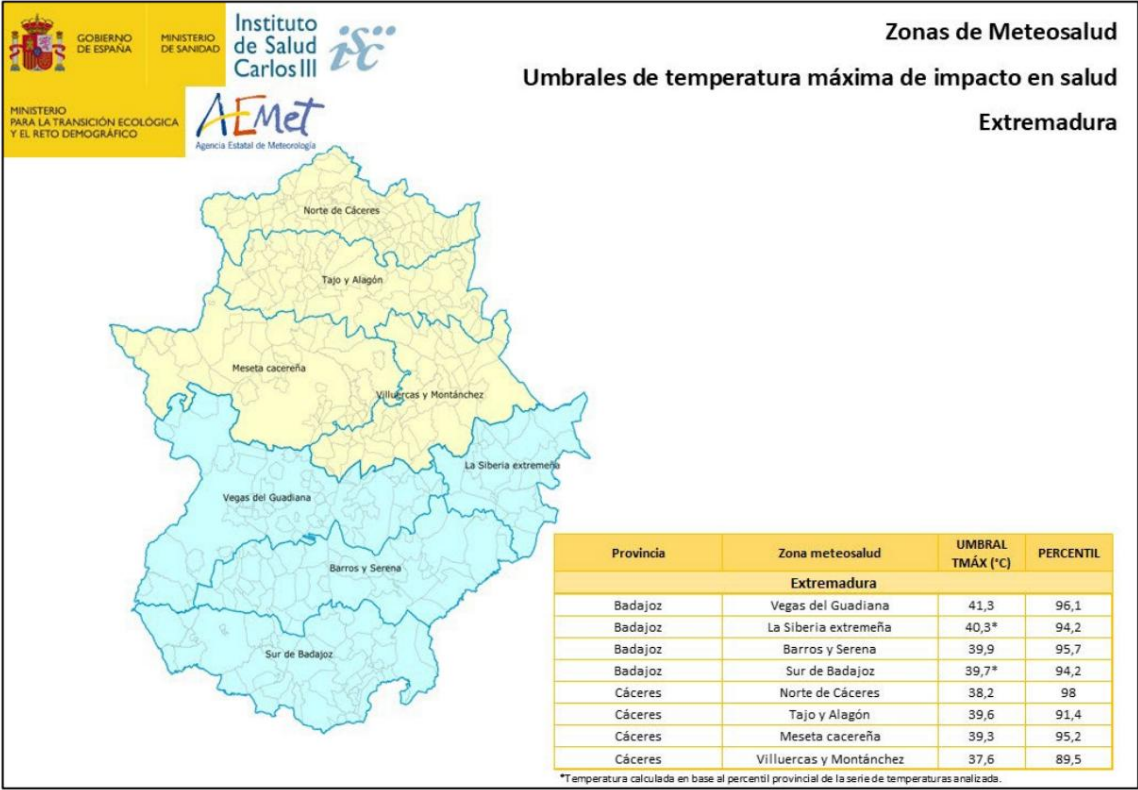


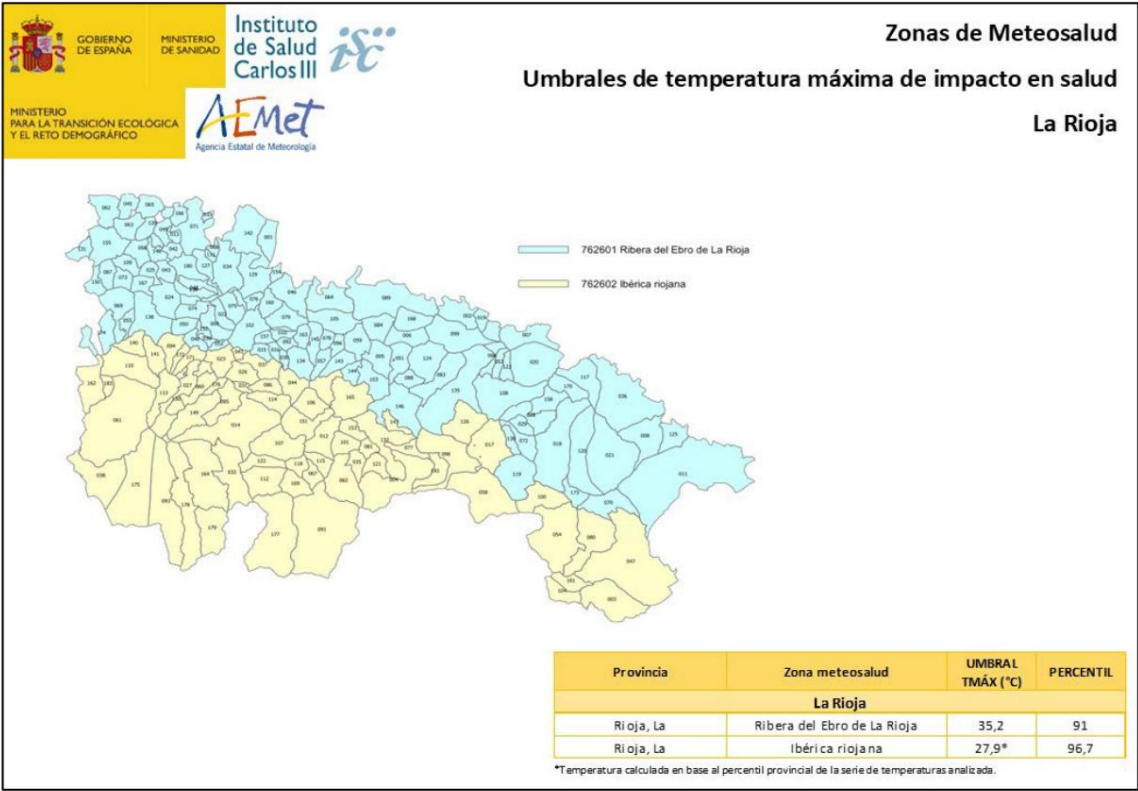
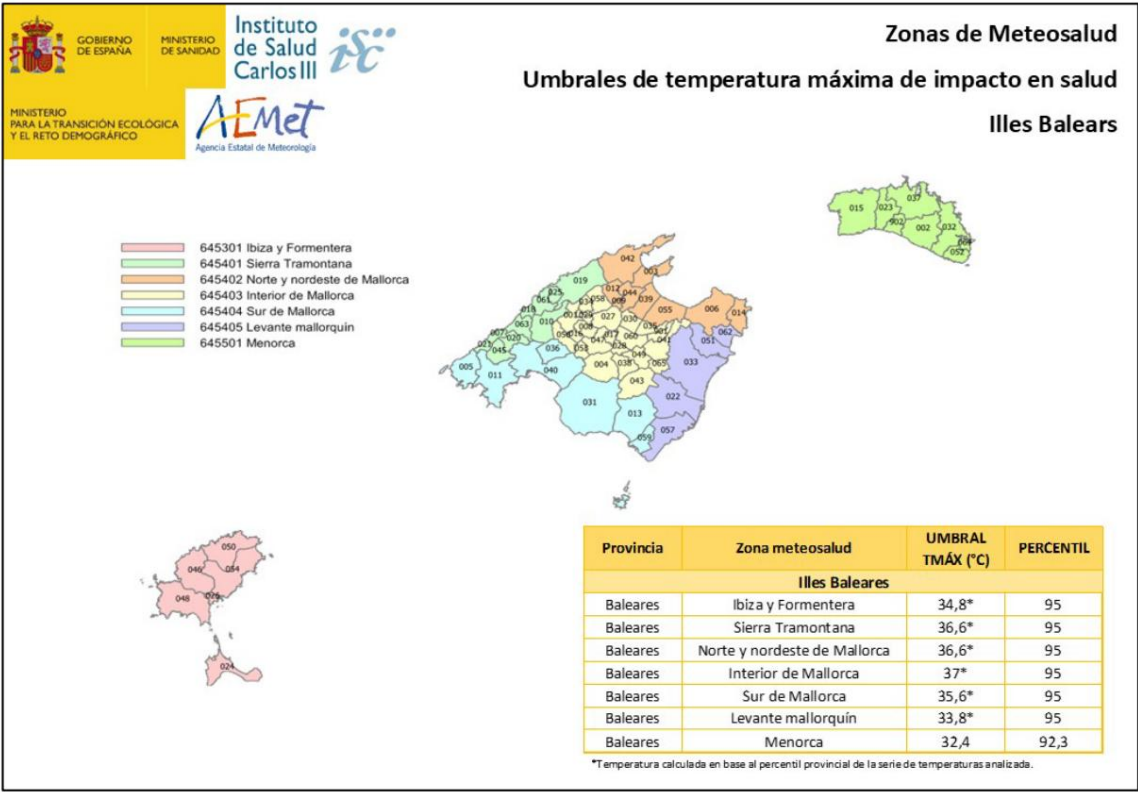


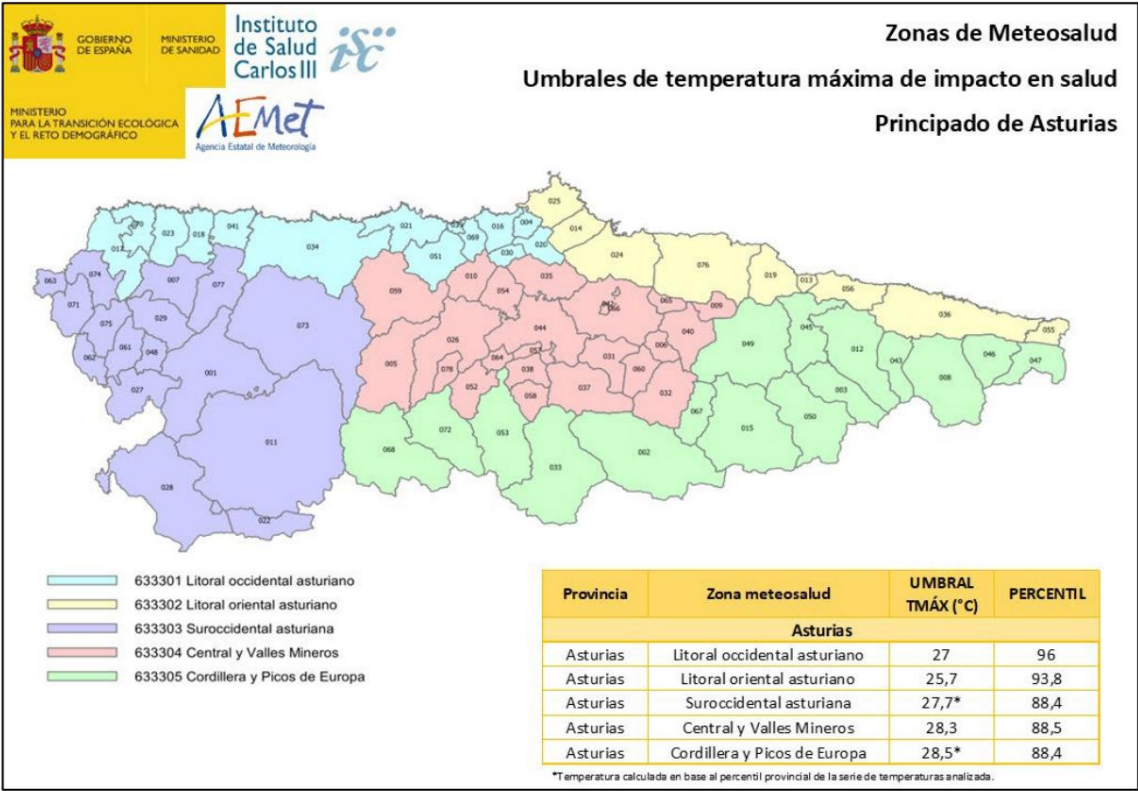
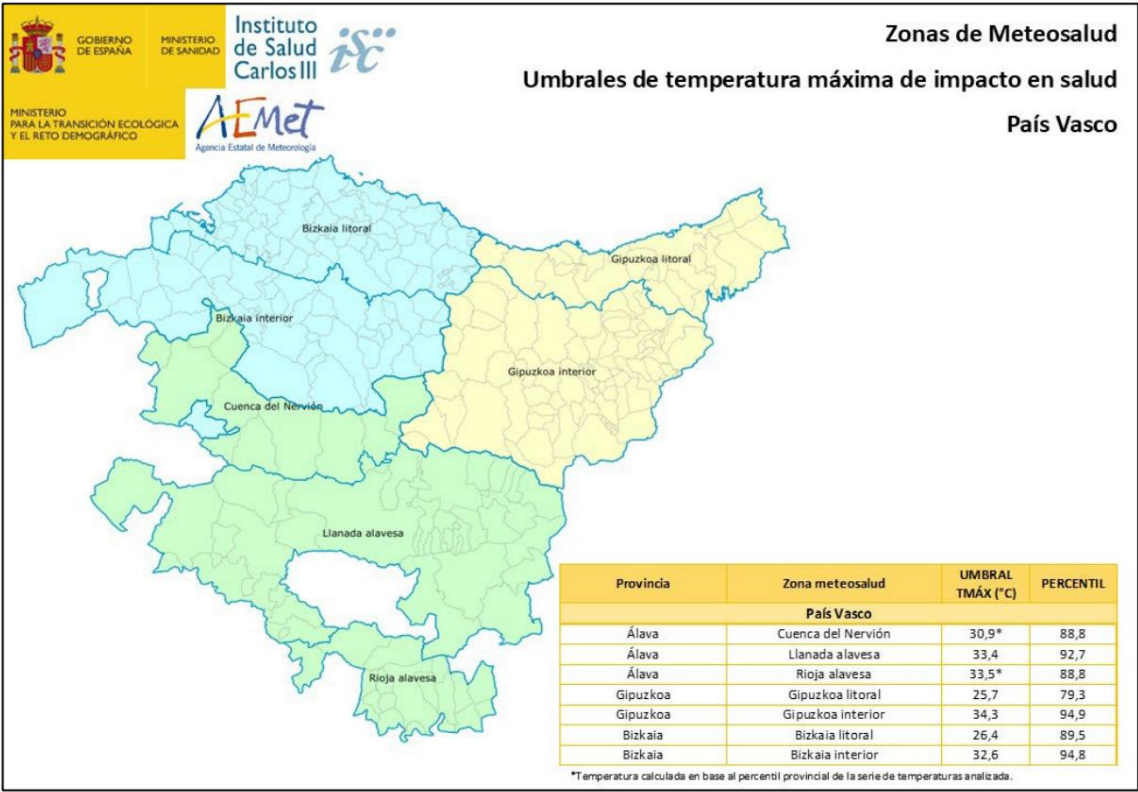


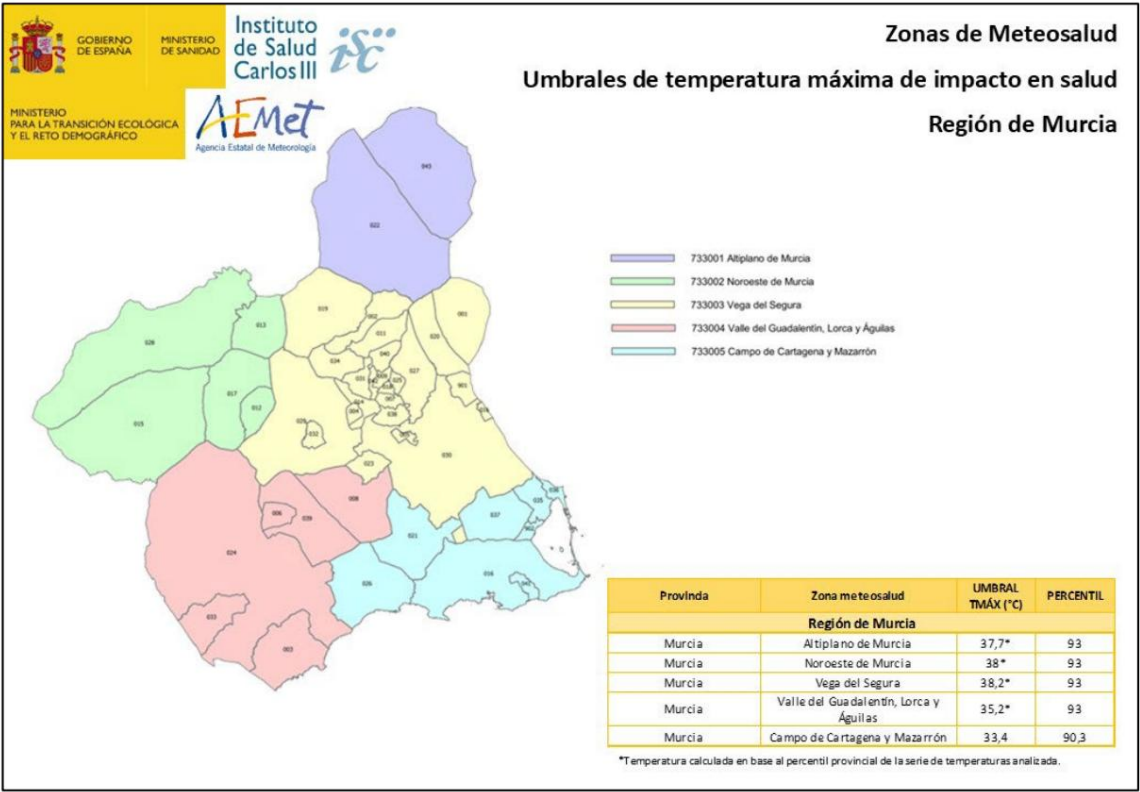












Annex II. Information relating to deaths from heat stroke

With the aim of improving the monitoring of mortality from heat stroke during the activation period of the National Plan of Preventive Actions against the Effects of Excessive Temperatures on Health, the necessary information on each death from heat stroke that occurs in the national territory and subject to communication by the Autonomous Communities and Cities to the General Subdirectorate of Environmental Health and Occupational Health of the Ministry of Health through the email (sgsasl2@sanidad.gob.es) is detailed below.

This information comprises three blocks of information:

1. Individual and Exposure Data: to report on basic individual variables and exposure to excessive temperatures. In this case, the environmental and social risk factors included in the National Plan are considered.

2.- Clinical-care data: to report on the circumstances of admission, the clinical risk factors included in the National Plan and the individual condition at admission measured by three conditions associated with heat pathology, namely: hyperthermia, dehydration and hyponatremia.

3.- Data relating to the death and the communication of the information

The following details the data corresponding to each of the blocks:

Individual and Exposure Data

• Age:

Sex :

• Profession:

• Place of residence:

• Location where symptoms begin:

Exposure⁹ presents a risk factor:

• Sí • No

If yes, please specify which one(s):

⁹ **Exposure:** Presence of environmental or social risk factors (for example, people who live alone, on the street and/or in unfavorable conditions, lack of air conditioning and homes that are difficult to cool), exposure to heat waves for work-related reasons (manual work outdoors or that require high contact with hot environments), sports (sports of high physical intensity) or leisure.

- People who live alone
- Homeless people
- People with unfavorable economic conditions
- People who live in collective residences
- Lack of air conditioning
- Housing difficult to cool
- Exposure due to work-related reasons (manual work outdoors or indoors and/or requiring high contact with hot environments)
- Sporting reasons (sports of high physical intensity)
- Leisure reasons
- Other:

Clinical and healthcare data

1. Presence of individual risk factors¹⁰.
• Sí • No
If yes, please indicate which one(s).
.....
2. Hospital admission location:
3. Date of hospital admission:
4. Hospital admission service (emergency, cardiology, geriatrics...):
.....
5. Presence upon entry of:
 - a. **Hyperthermia** • YES • NO
 - b. **Dehydration** • YES • NO
 - c. **Hiponatremia** • SI • NO

¹⁰ **Individual risk factors:** such as cardiovascular, respiratory and mental diseases (dementia, Parkinson, Alzheimer, ...); chronic diseases (diabetes mellitus, morbid obesity...); medical treatments (diuretics, neuroleptics, anticholinergics and tranquilizers); memory disorders, difficulties in understanding or orientation or little autonomy in daily life; acute illness during an episode of excessive temperature; consumption of alcohol and other drugs.

Death details and communication

1. Place of death:
2. Date of death:
3. Date of receipt of death information at the Service of the Autonomous Community or Autonomous City responsible for its transmission to the Deputy Directorate General for Environmental Health and Occupational Health of the Ministry of Health:
4. Date the information was sent to the Deputy Directorate General of Health Environmental and Occupational Health Department of the Ministry of Health:

Annex III. General recommendations to prevent the effects of high temperatures on health

General recommendations to prevent the effects of high temperatures about health

1. Drink water and other fluids frequently, even if you don't feel thirsty and independently.
of the physical activity you perform.
2. Avoid caffeinated, alcoholic or very sugary drinks, as they can promote dehydration.
3. Although anyone can suffer from a heat-related problem, pay special attention to: babies and children, infants and pregnant women, as well as the elderly or those with illnesses that can be aggravated by the heat (such as heart disease, kidney disease, diabetes, hypertension, obesity, cancer, conditions that impair mobility, dementia and other mental illnesses, as well as drug or alcohol abuse).
4. Stay in cool, shady or air-conditioned places as much as possible, and cool down whenever you need to.
5. Try to reduce physical activity and avoid outdoor sports in the midday hours.
6. Wear light, loose-fitting, breathable clothing.
7. Never leave anyone in a parked and locked vehicle (especially minors, the elderly, or people with chronic illnesses).
8. Consult your healthcare professional if you experience symptoms that last for more than one day.
time and that may be related to high temperatures.
9. Keep your medicines in a cool place; heat can alter their composition.
and its effects.
10. Eat light meals that help replenish the salts lost through sweat.
(salads, fruits, vegetables, juices, etc.).

Annex IV. Mirror Zones

Below are shown the meteosalud zones that presented excess missing values in the temperature records (>10% missing values), for which it was agreed to take as the threshold temperature of that meteosalud zone, the percentile of the province obtained from the time series of its meteorological mirror zone defined by AEMET.

Table 3. Relationship of zones with excess missing values in their temperature series and their meteorological mirror zone

Province	Weather Health Zone	Mirror area for temperature
Balearic Islands	Tramontana Mountains	North and northeast of Mallorca
Burgos	Treviño County	Llanada Alavesa
Ciudad Real	Northern Mountains and Widths	Guadiana Valley
Basin	Alcarria conquense	The La Mancha region of Cuenca
Granada	Guadix and Baza	Genil Basin
Guadalajara	Parameras de Molina	Guadalajara mountain range
Cantabria	Villaverde Center and Valley	Interior of Bizkaia
Toledo	Sierra de San Vicente	North of Cáceres

Annex V. Interpretation of the health risk levels for epidemiological heat waves

The National Plan defines four levels of health risk from high temperatures, identified by a traffic light-type color code: level 0 (absence of risk) and levels 1, 2 and 3, which indicate increasing health risk and whose colors are yellow, orange and red, respectively.

Extreme temperatures affect the health of all people, but the level of risk also depends on a number of personal, social, and environmental factors.

Among the possible risk factors are:

- **Personal:** Infants and children under 4 years old, people over 65 years old, pregnant women, people with cardiovascular, respiratory, or chronic diseases. People undergoing medical treatments. People with disorders
mental health issues, memory problems, difficulties in understanding or orientation, or limited autonomy in daily life, among others.
- **Environmental, work or social factors:** people who live alone, homeless people, people with unfavorable economic conditions, migrants and/or tourists, homes that are not well heated or without the ability to regulate the temperature, excessive exposure to heat for work, sports or leisure reasons (especially between 2:00 p.m. and 7:00 p.m.), among others.

The greater the number of risk factors, the greater the vulnerability. Among these, age is a significant factor, as mortality is concentrated primarily in those over 65, and especially in those over 75. Similarly, poverty, which prevents access to well-insulated housing and protection from high temperatures, is another crucial factor.

Taking this into account, the risk levels of the National Plan can be interpreted as follows:

A **yellow risk level** implies:

- Mild risk for people over 65 with other risk factors.

An **orange risk** level means

- Mild risk in the general population without risk factors
- Moderate risk for people over 65 years of age or under 65 years of age with multiple risk factors.
- High risk for those over 65 with other additional risk factors

A **red risk level** implies:

- Moderate risk in the general population without risk factors
- High risk for people over 65 and under 65 with any risk factor
- Extreme risk for those over 65 with other additional risk factors

The prolonged duration of red risk levels due to heat wave episodes represents an extreme risk level for the entire population, especially for people in a more vulnerable situation.

Credits

Interministerial Commission

Preside	Director General of Public Health and Health Equity. Ministry of Health
Vocal	Deputy Director General of Civil Protection and Emergencies. Home Office
Vocal	Deputy Director of the State Meteorological Agency. Ministry for Ecological Transition and the Demographic Challenge
Vocal	Deputy Director of the Institute for the Elderly and Social Services (IMSERSO). Ministry of Social Rights, Consumer Affairs and Agenda 2030
Vocal	Deputy Director General of Institutional Relations of the General State Administration in the Territory. Ministry of Territorial Policy and Democratic Memory
Secretariat	Deputy Director General of Environmental Health and Occupational Health. Ministry of Health

Interministerial Working Group

Preside	Deputy Director General of Environmental Health and Occupational Health.
	DG Civil Protection and Emergencies
	Institute for the Elderly and Social Services (IMSERSO)
	Ministry of Territorial Policy and Democratic Memory
	Carlos III Health Institute. ENS
	Carlos III Health Institute. CNE
	State Meteorological Agency (AEMET)
	Center for Coordination of Health Alerts and Emergencies (CCAES)
	Environmental Health and Occupational Health

Madrid, 2026